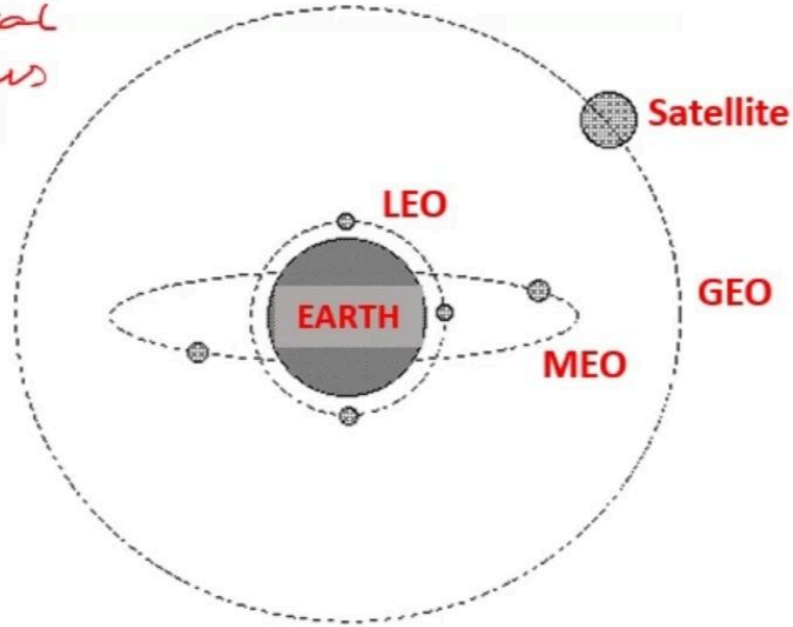
Introduction to Satellite Orbits

- An orbit is a regular, repeating path that an object in space takes around another one.
- An object moving around a planet in an orbit is called a satellite.
- The satellite can be rotated around the earth through different paths. These paths are known as orbits of the satellite.

An orbit is characterized by attributes:

- Altitude
- Inclination

Orbital radius



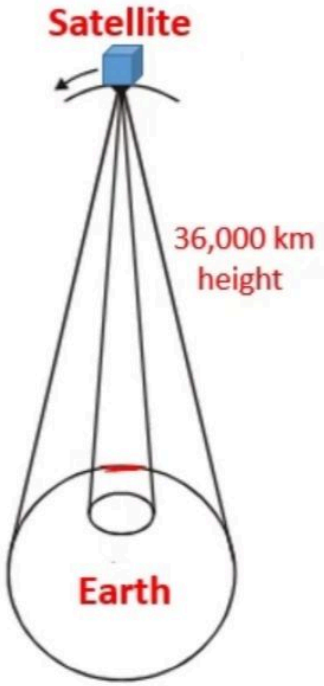
- The height of the satellite from the surface of the earth is known as altitude.
- Orbital inclination is the angle between the plane of an orbit and the equator.

Geostationary or Geosynchronous Earth Orbit (GEO)

- GEO satellites are synchronous with respect to earth. They appear to be stationary when viewed from earth.
- These satellites are placed in the space in such a way that three satellites are sufficient to provide connection throughout the surface of the Earth.
- The orbit of these satellites is circular. Lifetime expectancy of these satellites is 15 years.

There are three conditions which lead to geostationary satellites:

1) The satellite should be placed approximated to 36,000 km above the surface of the earth.
2) These satellites must travel with the same speed as of earth, and in the direction of motion of earth
3) The inclination of satellite with respect to earth must be 0^0



Advantages:

- It gives a large coverage area
- GEO satellites have a 24 hour view of a particular area
- These satellites are used for TV and radio broadcast, weather forecast, telephone networks etc.

Disadvantages:

- Satellite's distance causes weak signal and a time delay in the signal, which is a drawback for point to point communication.
- Satellites are centered above the equator therefore they have difficulty of broadcasting signals to near polar regions

Low Earth Orbit (LEO)

- Low Earth Orbit is relatively close to Earth's surface. It is normally at an altitude of 500-1500 km but could be as low as 160 km above Earth.
- In this highly inclined orbit. Each LEO satellite will only be visible from the earth for around ten minutes.
- LEO is commonly used for Satellite imaging, communication and remote sensing satellite systems etc.

Advantages:

- LEO satellite gives strong signal strength
- Less delay time
- Higher look angle
- Less path loss or beam spreading
- Easier to achieve high levels of frequency reuse

Disadvantages:

- Smaller area of coverage
- The need for many satellites for global coverage
- The short time of visibility with a high elevation requires additional mechanisms for connection handover between different satellites.
- Short lifetime of about five to eight years due to atmospheric drag and radiation

Medium Earth Orbit (MEO)

- Medium Earth orbit comprises a wide range of orbits anywhere between LEO and GEO.
- A MEO satellite is in orbit somewhere between 5,000 km and 10,000 km above the earth's surface.
- MEO satellites are similar to LEO satellites in functionality.
- These satellites are visible for much longer periods of time than LEO satellites, usually between 2 to 8 hours.
- They have a larger coverage area than LEO satellites.
- MEO is commonly used for navigation systems including the U.S. Global Positioning System (GPS).

Advantage:

- MEO satellite gives good signal strength
- Less delay time i.e. about 70-80 ms
- A MEO satellite's longer duration of visibility and wider footprint means less no. of satellites are required in a MEO network than a LEO network

Disadvantage:

- More number of satellites for global coverage
- Satellite's distance gives it a longer time delay and weaker signal than a LEO satellite
- The satellites need higher transmit power and special antennas for smaller footprints

Sun Synchronous Orbits (SSO)

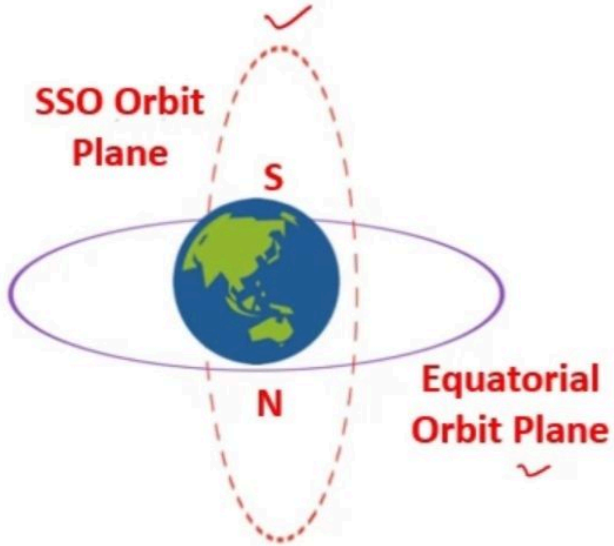
- Satellites in this orbit rise and set with the sun
- Satellites in this orbit always face the sun and hence they never go through an eclipse
- For these satellites, the surface illumination angle will be nearly the same every time
- The illumination angle of a any point of the Earth's surface is zero if the Sun is precisely overhead and it is 90^0 at sunset and at sunrise
- Earth resource satellites are launched in this orbit. Satellites can take earth's surface photographs because it is illuminated by sunlight

Special cases of the sun-synchronous orbit are the noon/midnight orbit

- Sun-synchronous orbit (SSO) is a particular kind of polar orbit
- These orbits are relevant for those countries which are located near the poles
- Polar orbits are a type of LEO as they are at low altitudes in between 200 to 1000 km

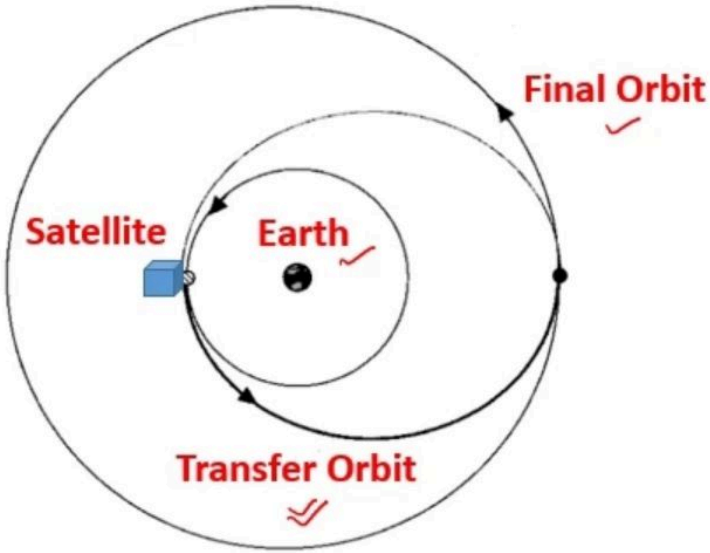
Surface illumination angle:

It is the angle between the inward surface normal and the direction of light.



Transfer Orbit and geostationary transfer orbit (GTO)

- There are many types of transfer orbits and they vary on the basis of their energy efficiency and speed of transfer
- Some of them are: Hohmann transfer orbit, an elliptical orbit which is used to transfer a spacecraft between two circular orbits of different altitudes
- This is an intermediate orbit having a highly eccentricity
- Generally it is used by GEO satellites to reach their final destination orbits



- This orbit is connected to the LEO orbit at the point of perigee forming a tangent
- It is also connected to the GEO orbit at the point of apogee again forming a tangent

Inclined Orbit

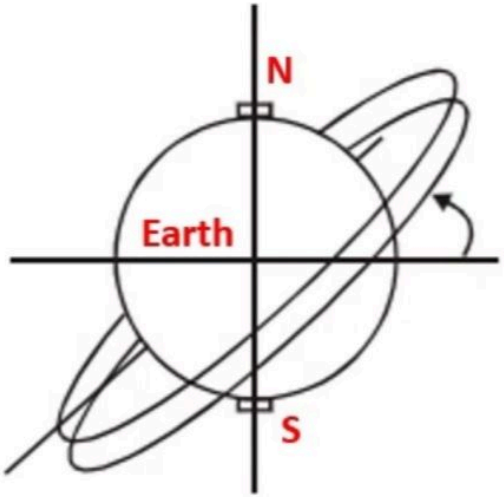
- The satellite following an orbital path other than equatorial or polar is known as Inclined Orbit
- A satellite is said to be in an inclined orbit around Earth if the orbit shows an angle other than 0° to the equatorial plane
- This angle is called the orbit's inclination angle

- Inclined orbits are used by many countries for their domestic communication
- The height of the inclined orbit is set to cover the area of interest
- The time for which satellite is visible from a point on the earth can also be controlled

Molniya Orbit

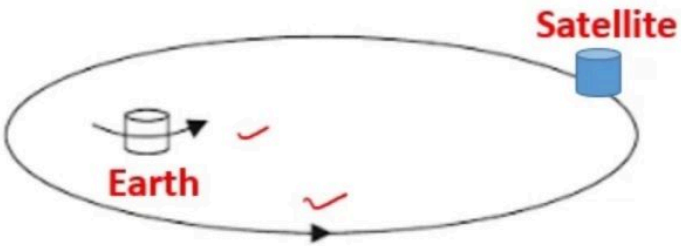
- It is highly inclined, highly eccentric, elliptical orbit with an orbital period of about 12 hours
- It has a perigee of 400km and an apogee of 40000km
- Its eccentricity is 0.75 and it has an orbit inclination of 65^0

- It is popular with Russia and its neighboring countries because it serves the purpose of geosynchronous orbit for high latitude regions
- Three satellites placed at different phases of the same Molniya orbit are capable to provide uninterrupted services
- The satellites in Molniya orbit are quite easy to launch in comparison to their geostationary counterparts



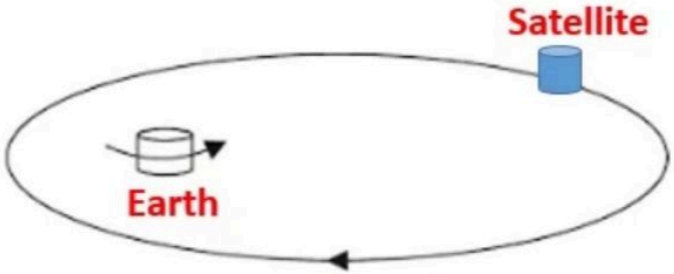
Prograde and Retrograde Orbit

- An orbit in which the satellite moves in the same direction of earth rotation is known as **Prograde Orbit**
- Inclination of this orbit is lies in between 0° and 90°
- It is also known as direct orbit
- Most of the satellites are launched in this orbit because earth's rotational velocity provides part of orbital velocity therefore saving in launch energy



Prograde Orbit

- An orbit in which the satellite moves in the opposite direction of earth rotation is known as **Retrograde Orbit**
- Inclination of this orbit is lies in between 90° and 180°
- Very few satellites are launched into this orbit because the quantity of fuel required to launch them is much greater in comparison to prograde orbit



Retrograde Orbit